The missing Copson–Cesàro embedding regime was completely characterized in 2022

Literature-status note for arXiv:1507.07866

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Status: literature_already_answered. The open regime $p_2 > q_2$ in the embedding $\mathrm{Cop}_{p_1,q_1}(u_1, v_1) \hookrightarrow \mathrm{Ces}_{p_2,q_2}(u_2, v_2)$ is covered by Gogatishvili, Pick, and Ünver, arXiv:2203.00596 / JFA 283 (2022). Their Theorem 2.1 gives necessary and sufficient conditions, with two-sided estimates of the optimal constant, for the equivalent four-weight Hardy–Copson inequality for all finite positive parameters. Remark 2.3 explicitly transfers the theorem back to the stated embedding.

The source problem

Gogatishvili, Mustafayev, and Ünver characterize

$$\mathrm{Cop}_{p_1,q_1}(u_1, v_1) \hookrightarrow \mathrm{Ces}_{p_2,q_2}(u_2, v_2)$$

for $0 < p_1, p_2, q_1, q_2 < \infty$ under $p_2 \le q_2$. In the Introduction they explain that their duality method works only in that range and state: "in the case when $p_2 > q_2$ the characterization of these embeddings remain[s] open" [1, p. 3]. Lemma 7.1 already disposes of $p_1 < p_2$: under the paper's nondegeneracy assumptions no such embedding can hold. Thus the substantive omitted range has $p_2 > q_2$ and $p_2 \le p_1$.

The later theorem and the exact bridge

The later paper studies the same inequality and makes the substitutions

$$r = \frac{p_2}{p_1}, \qquad q = \frac{q_2}{p_1}, \qquad p = \frac{q_1}{p_1}, \qquad u = u_2^{q_2}, \quad v = v_1^{-p_2} v_2^{p_2}, \quad w = u_1^{q_1}. \tag{1}$$

After replacing f by $(f v_1)^{p_1}$ and taking the p_1-th power, the embedding inequality is equivalent to

$$\left(\int_0^\infty \left(\int_0^t f(s)^r v(s)\, ds \right)^{q/r} u(t)\, dt \right)^{1/q} \le C \left(\int_0^\infty \left(\int_t^\infty f(s)\, ds \right)^p w(t)\, dt \right)^{1/p}, \tag{2}$$

with $C = c^{p_1}$ for the two optimal constants [2, pp. 5–6].

Theorem 2.1 of the later paper characterizes (2) for every $0 < r \le 1$ and every $0 < p, q < \infty$, in seven exhaustive parameter regions, and estimates its best constant in each region [2, pp. 6–8]. The old missing hypothesis $p_2 > q_2$ is exactly $q < r$. Within $r \le 1$, its possibilities are exhausted as follows:

$$p \le q < r \quad (\text{case (ii)}), \qquad q < p \le r \quad (\text{case (v)}), \qquad q < r < p \quad (\text{case (vi)}).$$

Remark 2.3 then states explicitly that the best constant in the original Copson-to-Cesàro embedding is obtained from Theorem 2.1 by the substitutions (1) [2, p. 8]. This is a full answer, not merely a sufficient condition or an answer for special weights.

Independent confirmation and scope

The authors' 2024 survey identifies the old $p_2 \leq q_2$ results and says that the complete characterization of the same embedding, "without any restrictions on parameters or weights," was given by the 2022 JFA paper [3, p. 4]. The conclusion here is therefore a literature-status identification; no new proof of the seven-case criterion is claimed.

The statement concerns finite positive exponents, exactly the parameter range of arXiv:1507.07866. The endpoint problem with infinite exponents is not needed to answer that source question.

References

[1] A. Gogatishvili, R. Mustafayev, and T. Ünver, *Embeddings between weighted Copson and Cesàro function spaces*, arXiv:1507.07866; Czechoslovak Math. J. **67** (2017), 1105–1132.

[2] A. Gogatishvili, L. Pick, and T. Ünver, *Weighted inequalities involving Hardy and Copson operators*, arXiv:2203.00596; J. Funct. Anal. **283** (2022), Paper 109719.

[3] A. Gogatishvili and T. Ünver, *On weighted Cesàro function spaces*, arXiv:2410.22301v2 (2025).